

Vowel Allophony Improves Maximum-Likelihood Classification of Warlpiri Consonants

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Abstract

Australian languages typically have a large number of place contrasts and relatively few manner contrasts. We investigated the acoustic cues that native speakers of Warlpiri, an Australian language with stops in five places of articulation, could use to distinguish them. Using a maximum-likelihood classifier, we show that acoustic cues intrinsic to consonants are not sufficient for this purpose. Adding information from consonant-extrinsic acoustic cues in adjacent vowels improved consonant classification for both CV and VC sequences. In both CVs and VCs, consonant classification benefited the most from allophony in formants at midpoints of adjacent vowels. We discuss the implications of our results and the potential of using maximum-likelihood classifiers for generating predictions based on probabilistic models of speech perception for low resource languages.

Index Terms: Warlpiri, consonants, place of articulation, acoustics, coarticulation, allophony, maximum-likelihood classification, confusion

1. Introduction

Australian languages are well known for having a large number of place distinctions in their consonants, particularly for stops [1, 2, 3, 4]. For instance, it is most common across the world's languages to have three place distinctions in stop consonants: bilabial, dental/alveolar, and velar. However, Australian languages typically have 4-6 place distinctions. These include bilabial, dental, alveolar, retroflex, palatal and velar contrasts. How do speakers of Australian languages distinguish a crowded place continuum with the high accuracy [5] observed in native listeners?

Cross-linguistically, many acoustic cues to stop place have been identified. These include cues intrinsic to the stop itself: stop duration [6]; spectral moments of the burst, i.e. center of gravity, standard deviation, skewness and kurtosis [7, 8]; and relative intensity of the stop burst with respect to the following vowel [9, 10]. The locus of the second formant has also been shown to be a primary descriptor of stop place of articulation [11]. Additionally, cues extrinsic to the stop, in particular second formant cues in adjacent vowels, are argued to be not only helpful but sufficient to distinguish consonant place [12].

However, there is an extensive body of research showing that the cues mentioned above are not equally effective in distinguishing the wider array of stop place distinctions in Australian languages. For instance, F2 locus and F2 transition into the vowel, often identified as a primary cue to stop place in other languages, has been reported to be insufficient to distinguish

between alveolars and retroflexes, and dentals and palatals, in Yanyuwa and Yindjibarndi [13, 14]. Small differences in articulatory place of articulation result in similar acoustic profiles for stop consonants, which Butcher and colleagues have argued is a consequence of speakers prioritising production of cues that signal consonant place over others [2, 1, 15, 4]. As a result, coarticulatory influences of the vowel on the consonant have been argued to be reduced, particularly in CV sequences.

1.1. Stop cues in Australian languages

Instead, based on extensive fieldwork, it has been argued that distinguishing stop place in Australian languages requires a more distributed constellation of acoustic cues. These include many intrinsic cues like spectral tilt and skewness, which have been shown to help differentiate both the velar and palatal in Pitjantjatjara [16] and velar and bilabial in several other Australian languages [2]. Duration of stops, in particular voice onset time, has also been reported to help identify place of articulation, differentiating retroflex and alveolar consonants for 10 speakers of several Australian languages [2]; apical coronals from laminals in Wubuy [17]; and velars from retroflexes and alveolars, bilabials from all coronals, and apicals from the palatals in Warlpiri [18]. Additional stop-intrinsic cues include E-M/H, a measure of spectral tilt quantifying the difference between spectral energy in high- and mid-band frequencies, which has been shown to differentiate alveolars and retroflexes in some prosodic contexts in Wubuy [17]. Finally, another intrinsic cue, F2 locus, has also been reported to distinguish bilabial and velar stops in several Australian languages [1].

Extrinsic cues like vowel formants also distinguish stop consonant place in Australian languages. For example, formant transitions in the F2 and F3 of VC sequences have been shown to vary between alveolars and retroflexes in Gupapuyngu, and a lower F3 for the retroflex may cue this contrast in CVs in careful speech [2]. Past work has also shown that alveolars and retroflexes in Wubuy are differentiated by F1 and F2 transitions in CV sequences and F3 transitions in VCs [19].

Overall, in Australian languages the cues that distinguish place of articulation are varied and it is unclear how listeners of a specific language integrate these cues to successfully distinguish native phoneme contrasts. Adding to the complexity is evidence that specific acoustic cues to consonant place can vary substantively in CV and VC sequences [1].

1.2. The current study

Our goal in this paper was to take a more systemic view of consonant perception, and crucially demonstrate the utility of maximum-likelihood classifiers as a tool to complement fieldwork in under-resourced languages. To do so we investigated

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how intrinsic and extrinsic cues jointly signal place of constriction in the Australian language Warlpiri. Warlpiri is a Pama-Nyungan language spoken in central Australia by approximately 2,600 speakers [20]. Phonologically, it is a prototypical Australian language, with stops at five places of articulation without a phonemic voicing contrast: /p/, /t/, /tʃ/, /c/ and /k/. It has three vowels, /i/, /a/ and /u/, which are contrasted in length.

To investigate how acoustic cues contribute to consonant place identification in Warlpiri, we trained a maximum-likelihood classifier over CV and VC sequences with intervocalic Cs on various combinations of acoustic cues. A maximum-likelihood classifier minimises categorisation error rate over a test set balanced across categories (either CV or VC sequences), presenting an upper bound on the performance of an optimal listener under Bayesian models that have been found to capture aspects of human speech perception [21, 22, 23]. That is, maximum-likelihood classifiers can provide a window into what is possible for an optimal human listener over the entire system of consonants.

In the baseline model, we used consonant-intrinsic cues proposed in the literature to distinguish the place of articulation for stop consonants. These include duration of the consonant [2], burst spectral moments, and F2 onset locus [2, 17, 16], and E-M/H [17]. All of these cues have been reported to be useful to distinguish at least one pair of stop consonants in an Australian language. We also used relative burst intensity, a cue that has been identified as useful in other languages of the world. Our results show that Warlpiri stops are quite confusable based on the acoustic information in the consonant alone.

Next, we evaluated the relative contribution of two extrinsic cues to reduce consonant place confusability in Warlpiri. The two cues we assessed were formant transitions and allophonic differences in neighboring vowel quality (provided by the estimate of the formant values at the vowel midpoint). The utility of formant transitions to signal differences in consonant place has been widely attested in many different languages, including those in Australia. The role of allophonic differences in the vowel itself to signal differences in consonant place is relatively understudied.

We know based on fieldwork in Bolivia, that subtle differences in consonant place, for example between velar and uvular consonants, is reinforced by Cochabamba Quechua speakers with a difference in height of adjacent vowels [24]. Gallagher also confirms that Cochabamba Quechua listeners use vowel height in the surrounding vowels to distinguish the velar-uvular place contrasts. Such facilitatory effects naturally emerge in Bayesian models of speech perception [21, 22, 23] if perception is optimised over sequences longer than a segment, allowing listeners to exploit conditional dependence of phonetic distributions on adjacent segments [25]. Consistent with predictions of Bayesian models of speech perception, listeners have been shown to use vowel quality cues to distinguish adjacent consonants based on long-term exposure with their native language [26] as well as based on short-term exposure in the lab [27].

We evaluated the relative contribution of the two extrinsic cues to improve consonant place classification in Warlpiri by augmenting the input to the maximum-likelihood classifier in one of two ways. We either added the formant transition cues or the vowel midpoint information to the baseline model with the consonant-intrinsic cues. By comparing the relative improvement in classification of consonants with the addition of each of the two extrinsic cues, we sought to identify the cues that listeners could exploit to effectively distinguish Warlpiri stop consonants. A relative improvement in consonant classification

with the addition of either formant transitions or vowel midpoint information or both would confirm whether these extrinsic cues to distinguish consonant place in Warlpiri can be exploited by listeners to distinguish place contrasts.

2. Methods

2.1. Corpus

Data were sourced from Carmel O’Shannessy’s field recordings from DoReCo [28], which contains semi-spontaneous monologic narratives from a structured story-telling elicitation task. The data were forced aligned using MAUS [29], and boundaries were hand-corrected by the archive (see [30] for processing pipeline). All audio files labelled as having “good” acoustic quality in the archive were kept, resulting in an analysis of 14 female speakers aged 25-50.

Because we were interested in differences in classification of consonants within CV versus VC sequences, we restricted ourselves to consonants drawn from VCV sequences that could be included in both VC and CV data sets. This resulted in an initial data set of 5,081 tokens for both CVs and VCs.

Segments orthographically transcribed as retroflex flaps were included in place of those transcribed as retroflex stops /t/. This was because a) there were very few (10 total) tokens of /t/ within VCVs, and b) the retroflex flap /ɬ/ and stop /t/ have been reported to neutralise in Warlpiri in intervocalic position [18]. Thus, the five consonants included in the analysis were /p/, /t/, /ɬ/, /c/, and /k/.

2.2. Acoustic cue extraction

We used Praat [31] to extract 31 acoustic cues from the corpus. These cues were split into four groups labelled as follows:

- **S:** consonant-intrinsic cues including stop-internal (duration and burst COG, standard deviation, skewness kurtosis, E-H/M, and intensity) and formant locus cues (F1-F4 at consonant/vowel boundary)
- **T:** formant transition cues (change in F1-F4 in the 25% of vowel adjacent to consonant)
- **M:** vowel formant cues (F1-F4 at vowel midpoint)

The same set of 7 stop-internal cues (a subset of S) were used for classification of both CV and VC sequences, while the remaining 24 cues (formant locus cues within S as well as T and M) were specific to either the preceding or following vowel. Thus, each classifier used at most 19 cues.

Although not all phonemic stops in the corpus include a burst due to lenition, it was not possible to manually check all tokens for presence of a burst, which MAUS [29] does not annotate. For consistency, extraction of spectral measures made use of burst locations automatically identified using AutoVOT [32] with its pretrained classifier. Spectral moments were extracted following the approach of [33] from a 0.0256 ms window centered on these bursts and E-H/M was calculated using high and low frequency bands of 3,500 Hz - 8,000 Hz and 1,250 Hz - 3,000 Hz respectively, following [17].

Not all acoustic cues could be extracted for all tokens due to errors while running AutoVOT [32] or failure of Praat [31] formant tracking to identify a formant. In the end, the data set for the CV-classifier included 4,755 tokens with all CV-relevant cues and the data set for the VC-classifier included 4,801 tokens with all VC-relevant cues.

2.3. Maximum-likelihood classifiers

Maximum-likelihood classifiers categorise a token x as whichever class y (among CV or VC sequences) maximises the likelihood of the acoustic values of x given sequence class y :

$$C(x) = \underset{y}{\operatorname{argmax}} p(x|y) \quad (1)$$

For a given acoustic cue set, likelihoods $p(x|y)$ were calculated based on a multivariate Gaussian paramaterised by the sample mean and covariance of the training data with respect to those cues. Maximum-likelihood classifiers were trained over all logical combinations of cue sets S, T, and M (see section 2.2) resulting in 7 CV classifier models and 7 VC classifier models.

Following common practice in evaluation of classification model performance [34], we ran a stratified 10-fold cross validation, calculating macro-averaged F-scores¹ for each fold. In addition to F-scores by CV or VC sequence, segment-specific F-scores were computed by treating a predicted label as correct as long as the consonant or vowel was correctly identified.

For significance testing of improvements in F-score between select pairs of models, a permutation test [35] was performed. For a given pair of models, a p -value was computed by comparing the mean difference between macro-averaged F-scores across folds against a null distribution of 10,000 permuted means. For instance, to assess significance of F-score improvement in a CV-classifier trained on cue set STM versus S, we compare the average difference $\bar{d}^{(STM,S)}$ in macro-F-score $F_{macro,i}$ across folds i , given below, against a null distribution:

$$\bar{d}^{(STM,S)} = \frac{1}{10} \sum_{i=1}^{10} \left(F_{macro,i}^{(STM)} - F_{macro,i}^{(S)} \right) \quad (2)$$

Each permuted mean of the null distribution was computed by resampling 10 within-fold F-score differences with replacement and randomly flipping the sign with probability 0.5 before averaging. We used Bonferroni's adjustment to control for type 1 errors, setting the new α threshold $p < 0.005$ ($= \frac{0.05}{10}$).

3. Results

3.1. Overview

Macro-averaged consonant and vowel F-score distributions for each cue set are plotted in Figure 1. A breakdown of the F-scores, including within-class F-scores by consonant and vowel, is presented in Table 1 for CVs in Table 2 for VCs.

Overall, the vowels were distinguished with the highest accuracy with just the vowel midpoint information (M F-score in CV 0.815 and VC 0.769), whereas the consonants were quite confusable based on just the stop-internal cues (S F-score in CV 0.654 and VC 0.647). Crucially, at least in CVs, there was greater improvement in consonant identification with the addition of vowel midpoint information compared to transition cues.

3.2. Vowels

Not only was the classification accuracy of vowels with midpoint information the highest, it was also remarkably stable as long as the classifier included the vowel midpoint information (the M, TM, SM, and STM models; which had a narrow vowel F-score range of 0.790-0.815 for CVs, 0.748-0.769 for VCs). There were no significant differences between any of these

¹ We use F, not F1, for F-score to avoid confusion with first formant.

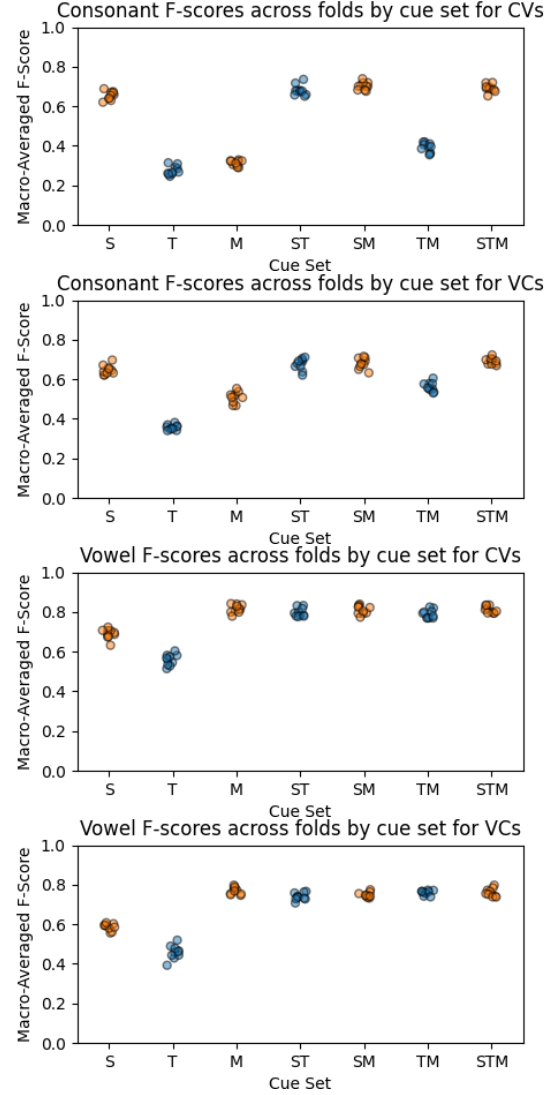


Figure 1: F -scores by cue set. S, T, and M denote stop-internal, transition, and vowel-midpoint cues (see Section 2.2).

models in CV sequences, when applying the Bonferroni adjustment ($p > 0.005$ for all comparisons). That is, vowel identification was unaffected by inclusion of stop-intrinsic or transition cues. This pattern broadly held true for the VC models as well, with the exception of a small difference found between the TM and M models, with transition information resulting in slightly worse performance ($\bar{d}^{(M,TM)} = 0.009$, $p = 0.0027$).

3.3. Consonants

For consonant identification in CV sequences, the addition of extrinsic cues, both transition information ($\bar{d}^{(ST,S)} = 0.026$, $p = 0.0015$) and vowel midpoint information ($\bar{d}^{(SM,S)} = 0.048$, $p = 0.0011$) significantly improved consonant classification. However, the magnitude of improvement in F-score from adding vowel midpoint cues, M, was almost twice that of adding transition cues, such that the SM model significantly outperformed the ST model ($\bar{d}^{(SM,ST)} = 0.021$, $p = 0.0015$).

Notably, the model that included both transition and mid-

Table 1: *F-Scores for CVs, sorted by macro-averaged consonant F-score*

Cue Set	By Consonant						By Vowel				By CV
	/p/	/t/	/tʃ/	/c/	/k/	Macro-Average	/i/	/a/	/u/	Macro-Average	Macro-Average
SM	0.605	0.660	0.745	0.814	0.686	0.702	0.780	0.848	0.803	0.810	0.558
STM	0.593	0.651	0.740	0.806	0.665	0.691	0.792	0.852	0.791	0.812	0.548
ST	0.595	0.624	0.731	0.797	0.656	0.681	0.771	0.837	0.786	0.798	0.540
S	0.584	0.575	0.702	0.770	0.641	0.654	0.649	0.732	0.687	0.689	0.473
TM	0.405	0.353	0.301	0.523	0.373	0.391	0.783	0.837	0.749	0.790	0.283
M	0.306	0.309	0.163	0.410	0.371	0.312	0.778	0.851	0.817	0.815	0.221
T	0.315	0.213	0.255	0.353	0.242	0.276	0.479	0.692	0.506	0.559	0.162

Table 2: *F-Scores for VCs, sorted by macro-averaged consonant F-score*

Cue Set	By Consonant						By Vowel				By VC
	/p/	/t/	/tʃ/	/c/	/k/	Macro-Average	/i/	/a/	/u/	Macro-Average	Macro-Average
STM	0.599	0.562	0.800	0.785	0.712	0.691	0.754	0.810	0.716	0.760	0.512
SM	0.615	0.544	0.784	0.771	0.690	0.681	0.728	0.810	0.705	0.748	0.502
ST	0.588	0.549	0.780	0.779	0.686	0.676	0.713	0.784	0.722	0.740	0.504
S	0.564	0.513	0.768	0.756	0.633	0.647	0.577	0.630	0.554	0.587	0.392
TM	0.429	0.396	0.676	0.686	0.619	0.561	0.721	0.824	0.734	0.760	0.425
M	0.315	0.349	0.624	0.676	0.579	0.508	0.722	0.837	0.749	0.769	0.376
T	0.319	0.236	0.405	0.366	0.456	0.356	0.399	0.579	0.396	0.458	0.177

point information in the CVs (STM F-score 0.691) actually performed worse than that with the midpoint alone (SM F-score 0.702) ($\bar{d}^{(STM,SM)} = 0.011$, $p = 0.0021$). This is consistent with the fact that the model that used transition information alone performed the worst of all models for CVs (T F-score 0.028). In sum, in CVs, the model with consonant intrinsic cues and vowel midpoint information (SM) had the highest consonant classification accuracy.

The consonant identification in VCs, although overall lower, showed a similar pattern. Again, there was a significant improvement in consonant classification compared to the S model (S F-score 0.647) with addition of either consonant-extrinsic cue: transitions (ST F-score 0.676, $\bar{d}^{(ST,S)} = 0.030$, $p = 0.0018$) or vowel midpoints (SM F-score 0.681, $\bar{d}^{(SM,S)} = 0.034$, $p = 0.0028$). Further, the SM model outperformed the TM model ($\bar{d}^{(SM,TM)} = 0.120$, $p = 0.001$). There was no significant difference in consonant F-score for the SM and STM models ($\bar{d}^{(SM,STM)} = 0.01$, $p = 0.03$).

4. Discussion

To our knowledge, the current study represents the first use of a probabilistic classification model to analyse how optimal use of available acoustic cues may facilitate perception of consonant place of articulation in an Australian language. Our results demonstrate the utility of using maximum-likelihood classifiers to investigate segment identification across the whole system, and show that these methodologies can be applied to low-resource languages using semi-spontaneous speech.

Using this approach we were able to classify both consonants and vowels using a large number of cues that have been reported to be useful in a variety of languages in Australia and outside. Further, using model comparison we were able to evaluate the utility of specific cue combinations. Our results show that provided only stop-intrinsic cues, the consonants of Walpiri

are quite confusable, as has been claimed broadly for Australian languages [2]. However, the addition of extrinsic cues, specifically from vowel midpoints, improves consonant classification significantly. We found this pattern of results for both CV and VC sequences. Further, we found that despite vowel allophony, there was no drop in classification accuracy of vowels.

We speculate that such vowel allophony is more likely to be useful to distinguish subtle place contrasts in languages with smaller inventories. As such, Walpiri with three vowels is an ideal test case. Whether we see differences in the utility of vowel allophony to distinguish consonant place contingent on the size of the inventory, predicted by McGahay’s Iterated Confusion Minimization model [25], we leave for future work.

One notable result was that the VC models performed slightly worse than CV models when intrinsic cues were included (all models that included S). Place identification of Cs benefited more from information in the midpoint of the following vowel. This is consistent with re-syllabification of the medial C with the following V in VCV sequences, as predicted by in-phase coupling within Articulatory Phonology [36, 37] or the Maximal Onset Principle within Optimality Theory [38]. Alternatively, it may simply reflect greater influence of the following vowel on C release bursts due to closer temporal proximity.

Although not presented here, we can use our maximum-likelihood classifier to generate confusion matrices based on acoustic overlap using any possible cue combination. Future work exploring the confusability of consonant pairs using more targeted combinations of acoustic cues may provide a benchmark to arbitrate between proposals of which cue combinations are most informative for native listeners. For instance, analysis not reported here shows that duration is particularly helpful among S (stop-internal) cues for distinguishing retroflex consonants, consistent with their realization as flaps with short duration. Future work might similarly apply more fine-grained analysis of individual cues to other consonants.

5. Generative AI Use Disclosure

Generative AI was not used in preparation of this paper.

6. References

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